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### User input deck for a gaming machine

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#### Abstract

A gaming machine for operating a wagering game comprises a user input deck and an emotive lighting structure. The user input deck includes a transmissive panel, a plurality of light sources, and an etched graphic. The plurality of light sources is constructed and arranged to transmit light into the transmissive panel via an edge thereof. The etched graphic is etched into a surface of the transmissive panel and is illuminated by the transmitted light. The emotive lighting structure is configured to present content related to the wagering game. The user input deck is disposed between the emotive lighting structure and a player position in front of the gaming machine. The emotive lighting structure extends below a height of the user input deck such that at least a portion of the content is visible through the transmissive panel.

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## **Background/Summary**

RELATED APPLICATIONS (1) This patent application claims priority benefit to U.S. Provisional Patent Application No. 63/402,198, filed Aug. 30, 2022. The disclosure of the 63/402,198 application is hereby incorporated by reference herein in its entirety.

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## FIELD OF THE INVENTION

(2) The present invention relates generally to gaming machines and methods of operating the same and, more particularly, relates to a user input deck for operating a gaming machine.

## BACKGROUND OF THE INVENTION

(3) The competitive landscape among gaming machine suppliers is fueled by accelerated product life cycles, increased product supply, continued entry of competitors, and increased investment in advanced development resources. To compete in the space of premium gaming machines with more expensive or extravagant components, suppliers have sought to differentiate such gaming machines with unique hardware configurations, game play mechanics, and third-party branded content. Such hardware configurations may include one or more large display devices in a landscape and/or portrait orientation, emotive lights surrounding the display devices, audio speakers that deliver a surround-sound audio experience, and a large user input deck with video and/or mechanical buttons for operating the game.

(4) As suppliers continue to offer a variety of premium gaming machines, a need exists for a premium gaming machine that further differentiates from the competition.

## SUMMARY OF THE INVENTION

(5) According to an embodiment of the present invention, a gaming machine for operating a wagering game comprises a user input deck and an emotive lighting structure. The user input deck includes a transmissive panel, a plurality of light sources, and an etched graphic. The plurality of light sources is constructed and arranged to transmit light into the transmissive panel via an edge thereof. The etched graphic is etched into a surface of the transmissive panel and is illuminated by the transmitted light. The emotive lighting structure is configured to present content related to the wagering game. The user input deck is disposed between the emotive lighting structure and a player position in front of the gaming machine. The emotive lighting structure extends below a height of the user input deck such that at least a portion of the content is visible through the transmissive panel.

(6) According to another embodiment of the present invention, a method of operating the gaming machine with the user input deck is disclosed herein.

(7) Additional aspects of the invention will be apparent to those of ordinary skill in the art in view of the detailed description of various embodiments, which is made with reference to the drawings, a brief description of which is provided below.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a gaming machine in operation according to an embodiment of the present invention.

(2) FIG. 2 is a front view of the gaming machine.

(3) FIG. 3 is a left side view of the gaming machine.

(4) FIGS. 4, 5, and 6 are close-up top perspective views of a user input deck of the gaming machine in operation, wherein the views showing different examples of LED-illuminated etched graphics in a transmissive panel of the user input deck.

(5) FIG. 7 is a close-up bottom perspective view of the user input deck of the gaming machine.

(6) FIG. 8 is an enlarged three-dimensional view of an LED strip inside a diffuser.

(7) While the invention is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail

herein. It should be understood, however, that the invention is not intended to be limited to the particular forms disclosed. Rather, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

#### DETAILED DESCRIPTION

(8) While this invention is susceptible of embodiment in many different forms, there is shown in the drawings and will herein be described in detail preferred embodiments of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to the embodiments illustrated. For purposes of the present detailed description, the singular includes the plural and vice versa (unless specifically disclaimed); the words “and” and “or” shall be both conjunctive and disjunctive; the word “all” means “any and all”; the word “any” means “any and all”; and the word “including” means “including without limitation.”

(9) For purposes of the present detailed description, the terms “wagering game,” “casino wagering game,” “gambling,” “slot game,” “casino game,” and the like include games in which a player places at risk a sum of money or other representation of value, whether or not redeemable for cash, on an event with an uncertain outcome, including without limitation those having some element of skill. In some embodiments, the wagering game involves wagers of real money, as found with typical land-based or online casino games. In other embodiments, the wagering game additionally, or alternatively, involves wagers of non-cash values, such as virtual currency, and therefore may be considered a social or casual game, such as would be typically available on a social networking web site, other web sites, across computer networks, or applications on mobile devices (e.g., phones, tablets, etc.). When provided in a social or casual game format, the wagering game may closely resemble a traditional casino game, or it may take another form that more closely resembles other types of social/casual games.

(10) The term “controller” may refer to a single physical controller or a set of controllers that operate in conjunction with each other and may or may not be located in the same enclosure inside the gaming machine. The types of controllers may include a central processing unit (CPU), a graphics processing unit, peripheral sub-controllers (e.g., an LED controller, a touchscreen controller, a value input device controller, a value output device controller, etc.), any combination of processors, etc. The controller may include specialized game-logic circuitry that has been approved by a gaming regulatory authority for deployment in the jurisdiction governed by that authority.

(11) Referring to FIGS. 1-3, there is shown a gaming machine **10** according to an embodiment of the present invention. The gaming machine **10** may be an electromechanical gaming terminal configured to play mechanical slots, or a video gaming terminal configured to play a video casino wagering game, such as slots, keno, poker, blackjack, roulette, craps, etc. The gaming machine **10** comprises a cabinet **12**, a video display device **14**, an interface panel **16**, an emotive trim lighting structure **18**, and a cantilevered user input deck **20**. The cabinet **12** extends from a bottom to a top of the gaming machine **10** (see FIG. 3) and may comprise one or more enclosures. The cabinet **12** contains a controller for operating the wagering game and managing peripherals of the gaming machine **10**. The cabinet **12** may also contain auxiliary equipment such as value input devices (e.g., bill/ticket acceptor) for funding the wagering game, value output devices (e.g., ticket printer/dispenser) for cashing out a game credit balance, and audio components (e.g., amplifier and subwoofer). The video display device **14** is mounted to the cabinet **12** by hooks, hinges, latches, fasteners, or the like. The video display device **14** may be flat, faceted, or curved and may be oriented in a landscape or portrait configuration. In the illustrated embodiment, the video display device **14** is a 75" flat, high-definition LCD or OLED oriented in a portrait configuration. The interface panel **16** is mounted to the cabinet **12** and disposed below the video display device **14** and behind the user input deck **20**. The interface panel **16** extends partially below a height/elevation of the user input deck **20**. The interface panel **16** carries a pair of audio speakers to deliver a stereo- or

surround-sound audio experience. Between the pair of audio speakers, the interface panel **16** may also carry a player-tracking display, a USB port, and ports for accepting and/or dispensing currency, tickets, and/or cards.

(12) The emotive lighting structure **18** is recessed relative to the video display device **14**. The emotive lighting structure **18** partially surrounds the video display device **14** and comprises a series of micro light-emitting-diode (LED) display panels bearing content related to the wagering game. The content may be active graphics and/or simple lighting. The series of micro-LED display panels includes 15 LED display panels along the left side of the video display device **14**, 15 LED display panels along the right side of the video display device **14**, and two LED display panels across the top of the video display device **14**. The two top LED display panels extend between the topmost LED display panels along the left and right sides of the video display device **14**.

(13) Referring to FIGS. **4** and **7**, the cantilevered user input deck **20** extends horizontally from the cabinet **12** (see FIG. **3**) towards a player seated in front of the gaming machine **10** such that the deck appears to be floating in air. The user input deck **20** is mounted to the cabinet **12** by hooks, hinges, latches, fasteners, or the like and is encompassed by a cast metal frame **21**. To power and communicate with any electronic components of the user input deck **20**, power and data cables may run between the user input deck **20** and the controller in the cabinet **12** via a rear connector **23**.

(14) The user input deck **20** comprises a video display device **22**, a pair of inductive mobile phone chargers **24** (see FIG. **4**, which also shows a player's mobile device on the left charger), a transmissive panel **26**, and a pair of electromechanical push-buttons **28a-b**. The pair of chargers **24** are disposed along a rear side of the video display device **22**. The transmissive panel **26** is composed of a glass or acrylic material, and the glass may be tempered. The term “transmissive” as used herein means that the panel is transparent or translucent, but not opaque, such that a player viewing the panel **26** from one side thereof (e.g., from the perspective shown in FIGS. **4-6**) can see through the panel **26** to the other side thereof. For example, as described below, content in the form of lit graphics or lighting from the portions of the emotive lighting structure **18** below the height of the panel **26** are visible to a player looking through the panel **26**. The video display device **22** is positioned in a U-shaped cutout in the transmissive panel **26**. In particular, the transmissive panel **26** includes left and right panel sections **26a-b** on opposite sides **22a-b** of the video display device **22** and a strip section **26c** extending between the panel sections **26a-b**. The strip section **26c** runs along a front side **22c** of the video display device **22**. The left and right panel sections **26a-b** enable the player, while playing the wagering game, to see through to the micro-LED display panels of the emotive lighting structure **18** which are below the height/elevation of the user input deck **20**. The LED display panels bear content, such as special effects, related to the wagering game. The panel sections **26a-b** have circular mounting apertures that contain the respective push-buttons **28a-b** such that the buttons appear to be suspended in air. The push-buttons **28a-b** are preferably located proximate a front edge of the user input deck **20** so that they can easily be reached and operated by the player. For a slot game, the player may press or “mash” either push-button **28a** or **28b** to initiate a spin of the slot reels.

(15) Although the user input deck **20** comprises the transmissive panel **26**, the deck **20** is constructed to support substantial weight (e.g., at least 300 pounds) at its outer edges and corners. In particular, the user input deck **20** includes a pair of cast metal ribs **30a-b** (see FIG. **7**) extending between the video display device **22** and the respective front left and right corners of the frame **21**. Electrical wiring may run through the ribs **30a-b** to power the push-buttons **28a-b** and transmit control signals between the controller and the buttons. The ribs **30a-b** are constructed and arranged to minimize impact on the player's sightlines through the transmissive panel **26** to the portions of the emotive lighting structure **18** below the height/elevation of the user input deck **20**.

(16) In the illustrated embodiment, the video display device **22** is a 24" flat, high-definition LCD or OLED oriented in a landscape configuration. The video display device **22** is outfitted with a touchscreen and bears touch keys enabling the player to operate the wagering game, including but

not limited to selecting different wager amounts, selecting different numbers of active paylines (for slot games with paylines), bringing up help and pays, calling a service attendant, and cashing out the player's balance on a game credit meter. The video display device **22** may also present at least a portion of an outcome of the wagering game.

(17) To attract players to the gaming machine **10** and further differentiate it from other premium machines on a casino floor, the user input deck **20** includes an LED lighting structure. The LED lighting structure comprises a continuous U-shaped flexible strip **32** of red, blue, and green (RGB) LED's and an etched graphic **34**. The LED strip **32** is mounted in a continuous channel that runs along the three sides **22a-c** of the video display device **22**. The LED's of the strip **32** may face the transmissive panel **26** to transmit light directly into the interior of the transmissive panel **26** via an adjacent edge of the panel **26**. Instead of transmitting light directly into the interior of the transmissive panel **26**, the LED's of the strip **32** may instead direct light onto a reflective or refractive surface that, in turn, reflects or refracts the received light into the interior of the transmissive panel **26**. To diffuse the light entering the transmissive panel **26**, the LED strip **32** may be enclosed or covered with a diffuser, or the adjacent edge of the panel **26** through which the light enters the panel **26** may be frosted. Referring briefly to FIG. **8**, there is shown an enlarged three-dimensional view of an exemplary flexible LED strip **32** covered by a diffuser **36** and mounted in a channel **38**. The channel **38** may be formed, for example, within the sides **22a-c** of the video display device **22**, or may be formed by an aluminum bar(s) **40** that, in turn, is inserted into a channel formed within the sides **22a-c** of the video display device **22**.

(18) Returning to FIGS. **4** and **7**, to transmit and contain the light received from the LED's, the upper and lower surfaces of the transmissive panel **26** are smooth and polished (except as described below). The transmissive panel **26** may have a thickness ranging from 3 mm to 5 mm. A second unitary transmissive panel (not shown) optionally overlays both the transmissive panel **26** and the video display device **22** to create a seamless upper surface to the user input deck **20**. This second panel likewise contains circular apertures, aligned with the mounting apertures in the transmissive panel **26**, to expose the push-buttons **28a-b**.

(19) To generate a graphic with the LED's of the strip **32**, the lower/bottom surface of the transmissive panel **26** is etched with the desired graphic **34** using surface etching, carving, and/or shading techniques. As light from the LED's propagates through the transmissive panel **26**, the etched graphic **34** catches and diffracts the light outwards, thereby illuminating the etched graphic **34** while the rest of the transmissive panel remains more uniformly transparent or translucent. In FIG. **4**, the LED-illuminated etched graphic **34** comprises a pattern of round dots that radiate outwardly from the sides **22a-c** of the video display device **22** and fade near the push-buttons **28a-b** and the front portion of the frame **21**. In FIG. **5**, the LED-illuminated etched graphic **34** comprises a pattern of diamonds that radiate outwardly from the sides **22a-c** of the video display device. The size of the diamonds decreases, and the spacing of the diamonds increases, as the diamonds get farther away from the sides **22a-b**. In FIG. **6**, the LED-illuminated etched graphic **34** comprises streaks of varying lengths that emanate horizontally from the sides **22a-b** of the video display device. The etched graphic **34** may have many other designs. The RGB LED's of the strip **32** are capable of producing millions of hues of light and are individually addressable. Via an LED sub-controller, the controller may control the LED's in different states of the wagering game to have different lighting modes, such as waves, sequential, slow glow, chasing/flashing, fading slowly, twinkle/flashing, and steady-on. The chosen mode may be synchronized with the current audio and/or video content presented in the wagering game. The controller may cause the LED's to illuminate a color(s) that complements or matches the dominant color(s) displayed on the video display device **22** and/or the video display device **14**. If the etched graphic **34** occupies central portions of the transmissive panel sections **26a-b**, that portion of the etched graphic **34** preferably is of low enough density to minimize interference with the player's sightlines through the transmissive panel **26** to the portions of the emotive lighting structure **18** below the

height/elevation of the user input deck 20.

(20) The gaming machine 10, including but not limited to the cabinet 12, the video display device 14, the interface panel 16, the emotive lighting structure 18, the user input deck 20, and their relative positioning, may use alternative designs to achieve the same functionality described herein. For example, while maintaining the transmissive panel 26 and LED lighting structure in the user input deck 20, the shapes, locations, quantities, and types of those and other components in the user input deck 20 may be varied from what is illustrated in the drawings. Instead of, or in addition to, residing along the sides of the video display device 22, the LED lighting structure including the LED strip 32) may reside along an inside edge of the frame 21 (except for the portion of the frame 21 behind the video display device 22) and be constructed and arranged to transmit light into the transmissive panel 26 via an adjacent outer edge thereof. The etched graphic 34 may radiate inwardly from the frame 21 instead of outwardly from the video display device 22.

(21) Each of these embodiments and obvious variations thereof is contemplated as falling within the spirit and scope of the claimed invention, which is set forth in the following claims. Moreover, the present concepts expressly include any and all combinations and subcombinations of the preceding elements and aspects.

## Claims

1. A gaming machine for operating a wagering game, comprising: a user input deck including a transmissive panel, a plurality of light sources, and an etched graphic, the plurality of light sources being constructed and arranged to transmit light into the transmissive panel via an edge thereof, the etched graphic being etched into a surface of the transmissive panel and being illuminated by the transmitted light; and an emotive lighting structure configured to present content related to the wagering game, the user input deck disposed between the emotive lighting structure and a player position in front of the gaming machine, the emotive lighting structure extending below a height of the user input deck such that at least a portion of the content is visible through the transmissive panel while the etched graphic is illuminated; wherein the user input deck includes a display device, a frame, and one or more ribs, the frame encompassing the transmissive panel, the one or more ribs being connected to both the frame and the display device and being immediately below the transmissive panel, and wherein the user input deck includes an electromechanical push-button within the transmissive panel and along one of the one or more ribs, the one of the one or more ribs carrying electrical wiring to the push-button.
2. The gaming machine of claim 1, further including a second display device configured to present the wagering game, the emotive lighting structure at least partially surrounding the second display device and including a series of micro light-emitting-diode (LED) display panels.
3. The gaming machine of claim 2, further including a cabinet, wherein the second display device, the emotive lighting structure, and the user input deck are mounted to the cabinet, the user input deck extending horizontally from the cabinet towards the player position.
4. The gaming machine of claim 1, wherein the plurality of light sources includes a strip of light-emitting diodes (LED's).
5. The gaming machine of claim 4, wherein the strip of LED's are disposed along one or more sides of the display device, the etched graphic radiating outwardly from the one or more sides of the display device.
6. The gaming machine of claim 5, wherein the LED's of the strip face the transmissive panel.
7. The gaming machine of claim 1, wherein the user input deck includes a diffuser positioned to diffuse the light before it enters the transmissive panel.
8. A user input deck for a gaming machine, the gaming machine including an emotive lighting structure configured to present content related to a wagering game, the user input deck disposed between the emotive lighting structure and a player position in front of the gaming machine, the

emotive lighting structure extending below a height of the user input deck, the user input deck comprising: a transmissive panel; a plurality of light sources constructed and arranged to transmit light into the transmissive panel via an edge thereof, a graphic etched into a surface of the transmissive panel and being illuminated by the transmitted light, at least a portion of the content on the emotive lighting structure being visible through the transmissive panel while the etched graphic is illuminated; a display device, a frame, and one or more ribs, the frame encompassing the transmissive panel, the one or more ribs being connected to both the frame and the display device and being immediately below the transmissive panel; and an electromechanical push-button within the transmissive panel and along one of the one or more ribs, the one of the one or more ribs carrying electrical wiring to the push-button.

9. The user input deck of claim 8, wherein the user input deck is mounted to a cabinet of the gaming machine and extends horizontally from the cabinet towards the player position.

10. The user input deck of claim 8, wherein the plurality of light sources includes a strip of light-emitting diodes (LED's).

11. The user input deck of claim 10, wherein the strip of LED's are disposed along one or more sides of the display device, the etched graphic radiating outwardly from the one or more sides of the display device.

12. The user input deck of claim 11, wherein the LED's of the strip face the transmissive panel.

13. The user input deck of claim 8, further including a diffuser positioned to diffuse the light before it enters the transmissive panel.

14. A method of operating a gaming machine, the gaming machine including a user input deck and an emotive lighting structure, the user input deck including a transmissive panel, a plurality of light sources, and an etched graphic, the etched graphic being etched into a surface of the transmissive panel, the user input deck disposed between the emotive lighting structure and a player position in front of the gaming machine, the emotive lighting structure extending below a height of the user input deck, the user input deck including a display device, a frame, and one or more ribs, the frame encompassing the transmissive panel, the one or more ribs being connected to both the frame and the display device and being immediately below the transmissive panel, the user input deck including an electromechanical push-button within the transmissive panel and along one of the one or more ribs, the one of the one or more ribs carrying electrical wiring to the push-button, the method comprising the operations of: presenting content on the emotive lighting structure; and transmitting light from the plurality of light sources into the transmissive panel via an edge thereof such that the etched graphic is illuminated by the transmitted light, at least a portion of the content being visible through the transmissive panel while the etched graphic is illuminated.

15. The method of claim 14, wherein the plurality of lights sources includes a strip of light-emitting diodes (LED's) disposed along one or more sides of the display device, the etched graphic radiating outwardly from the one or more sides of the display device.

16. The method of claim 14, wherein the transmitting operation includes diffusing the light with a diffuser before it enters the transmissive panel.

17. The method of claim 14, further including initiating, via the electromechanical push-button within the transmissive panel, a play of a wagering game at the gaming machine.

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